

MKT-10: Market Research Data

Project AEGIS / Carrington Storm Motors

Compiled Market Intelligence — Historical Data, Industry Forecasts, Adoption Models

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EXECUTIVE SUMMARY

This document compiles the market research data underlying CSM’s market sizing, competitive analysis, and go-to-market strategy. Data is organized by category: historical EMP/CME event impact data, grid infrastructure investment forecasts, residential construction market data, maritime fleet data, insurance market data, government spending data, consumer awareness data, and technology adoption models. All data points are cited to the extent possible; where citations are unavailable, the document notes the estimation methodology.

SECTION 1: HISTORICAL EMP/CME EVENT IMPACT DATA

1.1 The 1859 Carrington Event

Parameter	Value	Source
Date	September 1-2, 1859	Royal Greenwich Observatory records
CME transit time (Sun to Earth)	17.6 hours (typical is 2-4 days)	Carrington, R.C., Monthly Notices of the Royal Astronomical Society, 1859
Geomagnetic storm intensity	Estimated Dst = -1,760 nT (vs. -589 nT for March 1989)	Tsurutani et al., Journal of Geophysical Research, 2003
Kp index equivalent	9+ (off-scale; modern equipment saturation at Kp=9)	Cliver & Svalgaard, Solar Physics, 2004
Telegraph system impact	Global telegraph network disabled; operators received shocks; fires at multiple telegraph stations	Loomis, E., American Journal of Science, 1861; multiple contemporary newspaper accounts
Aurora visibility	Observed as far south as Cuba, Mexico, Hawaii, and northern Australia	Green & Boardsen, Advances in Space Research, 2006
Estimated recurrence interval	~150 years (meaning we are overdue based on statistical return period)	Riley, P., Space Weather, 2012: estimated 12% probability per decade of a Carrington-class event

1.2 March 1989 Hydro-Québec Blackout

Parameter	Value	Source
Date	March 13, 1989	Hydro-Québec operational records
Storm intensity	Dst = -589 nT; Kp = 9	NOAA SWPC historical records
Cause	GIC (geomagnetically induced currents) saturated transformers on Hydro-Québec’s 735 kV La Grande network	Kappenman, J., IEEE Spectrum, 1990
Duration of blackout	9 hours	Hydro-Québec
Population affected	6 million people in Québec	Hydro-Québec

Parameter	Value	Source
Economic impact	\$13.4 million direct cost (1989 dollars; ~\$30 million in 2026 dollars)	Bolduc, L., IEEE Transactions on Power Delivery, 2002
Equipment damage	One static VAR compensator destroyed; multiple transformer thermal alarms triggered	Kappenman, 1990
Recovery time	Full system restoration in ~9 hours; pre-emptive grid separation prevented wider cascading	Hydro-Québec

1.3 October-November 2003 Halloween Storms

Parameter	Value	Source
Dates	October 28 - November 4, 2003	NOAA SWPC
Solar source	AR 10486, one of the largest sunspot groups ever recorded	NOAA SWPC
Storm intensity	Dst = -383 nT (Oct 30); Dst = -401 nT (Nov 20)	NOAA SWPC
Grid impacts	Transformer failures in South Africa (15 transformers damaged/destroyed); Swedish Malmö blackout (50,000 customers, 1 hour); GPS navigation disruption; satellite anomalies (ADCS anomalies on multiple spacecraft)	Pulkkinen et al., Space Weather, 2005; Gaunt & Coetzee, IEEE PES, 2007
Aviation impact	FAA WAAS (Wide Area Augmentation System) unavailable for 30 hours; GPS-based navigation degraded	NOAA SWPC
Estimated economic impact	\$4-10 billion (global, including satellite operations, power grid, aviation, and GPS disruption)	Schulte in den Bäumen et al., Space Weather, 2014

1.4 May 2024 Gannon Storm (G5 Event)

Parameter	Value	Source
Date	May 10-13, 2024	NOAA SWPC
Classification	G5 (Extreme) — first G5 since 2003	NOAA SWPC
Kp index	Kp = 9 (maximum)	NOAA SWPC
Aurora visibility	Observed as far south as Florida, Mexico, and India	Multiple citizen science reports
Grid impacts	Minor grid irregularities reported in US and Canada; no major transformer damage; NERC GMD operating plans activated	NERC preliminary report, June 2024
Satellite impacts	Starlink reported degraded service; multiple satellite operators reported attitude control anomalies	SpaceX public statements; satellite operator reports

Parameter	Value	Source
GPS impacts	Precision agriculture GPS guidance degraded for 12-18 hours in North America; John Deere and other precision ag systems affected	NOAA SWPC; agricultural industry reports
Insurance impact	No major property insurance claims; event served as wake-up call for insurance industry — “near miss” that wasn’t Carrington-class but demonstrated vulnerability	Swiss Re sigma 02/2025 discussion
CSM significance	This event demonstrated to utilities, insurers, and governments that G5 events occur and cause measurable impact. The “imaginary threat” is now empirically demonstrated.	CSM analysis

1.5 Known EMP Events (Man-Made)

Event	Date	Details	Source
Starfish Prime (US nuclear test)	July 9, 1962	1.4 MT nuclear detonation at 400 km altitude over Johnston Atoll; EMP effects observed 1,445 km away in Hawaii (street lights failed, burglar alarms triggered, radio communications disrupted)	DOD historical records; declassified DTRA reports
Soviet Test 184 (Kazakhstan)	October 22, 1962	300 kT detonation at 290 km altitude; EMP effects damaged power lines and telephone systems 600 km from detonation point	Russian Federation declassified records; EMP Commission, 2008
Lightning EMP analogs	Daily occurrence	Natural lightning produces localized EMP effects; approximately 25 million cloud-to-ground strikes per year in the US alone (NOAA NSSL). While not comparable to HEMP, these demonstrate the vulnerability of unshielded electronics to EM transients.	NOAA National Severe Storms Laboratory

SECTION 2: GRID INFRASTRUCTURE INVESTMENT DATA

2.1 Global Grid Investment

Metric	Value	Source
Global grid investment (2023)	\$330 billion	IEA World Energy Investment 2024
Transmission investment (2023)	\$120 billion (36% of total)	IEA WEI 2024
Distribution investment (2023)	\$210 billion (64% of total)	IEA WEI 2024
Projected annual grid investment (2030)	\$500-600 billion	IEA Net Zero Roadmap, 2023 update
Projected annual grid investment (2035)	\$650-800 billion	IEA WEI 2024
Cumulative grid investment (2024-2035)	\$7-9 trillion	IEA WEI 2024
US grid investment (2023)	\$98 billion	EIA Annual Energy Outlook 2025
EU grid investment (2023)	€65 billion (~\$71 billion)	ENTSO-E Ten-Year Network Development Plan 2024
China grid investment (2023)	¥560 billion (~\$78 billion)	China Electricity Council

2.2 US Grid Asset Inventory

Asset	Quantity	Replacement Value	Source
High-voltage transmission lines (>230 kV)	257,000 circuit-kilometers	\$500 billion	EIA; NERC
Distribution lines	5.5 million miles (8.9 million km)	\$900 billion	EIA; ASCE
Substations (all voltages)	~55,000	\$800 billion	EIA; EPRI
Large Power Transformers (LPTs)	~2,000 (HV/EHV)	\$10-30 billion	DOE LPT Study, 2024
Distribution transformers	~60 million	\$180 billion	DOE; EEI
Circuit breakers (transmission)	~80,000	\$50 billion	NERC; EPRI
Control systems (SCADA, EMS)	~3,000 control centers	\$25 billion	DHS CISA; EPRI

2.3 Grid Vulnerability to GMD/EMP

Finding	Source
A severe GMD event (100-year benchmark) could damage 300+ EHV transformers in the US simultaneously	Kappenman, J., “Geomagnetic Storms and Their Impacts on the U.S. Power Grid,” Metatech Corp. for ORNL, 2010
Mean time to manufacture and replace 300 EHV transformers: 12-24 months each; global manufacturing capacity ~100 EHV transformers/year	DOE Large Power Transformer Study, 2017; updated 2024
E3 pulse from HEMP produces GIC effects similar to CME but more geographically concentrated (continental-scale vs. global)	EMP Commission, 2008
NERC identified 2,847 critical substations where GMD-induced failure would cause cascading outages affecting >500,000 customers each	NERC 2024 Long-Term Reliability Assessment
EPRI estimated average cost of \$1.2-\$3.8 million per HV substation for GMD protection	EPRI GMD Mitigation Report, 2023

SECTION 3: RESIDENTIAL CONSTRUCTION MARKET DATA

3.1 US Housing Stock

Metric	Value	Source
Total US housing units	144 million	US Census Bureau, American Housing Survey 2025
Single-family detached	92 million (64%)	AHS 2025
Owner-occupied	94 million (65%)	AHS 2025
Annual new residential construction	1.4 million units (2024)	US Census Bureau, New Residential Construction
Average new single-family home construction cost	\$392,241	NAHB, 2024
Average electrical system cost (new construction)	\$16,000-\$22,000	NAHB; RSMeans
High-value homes (>\$1M)	8.2 million	AHS 2025
Ultra-high-value homes (>\$5M)	~600,000	Coldwell Banker Global Luxury Report 2024

3.2 Luxury Home Market Characteristics

Finding	Source
37% of luxury home buyers cite “resilience features” as top-3 priority in new construction	Coldwell Banker Global Luxury Report 2024
Average luxury home electronics investment: \$50,000-\$250,000 (home automation, theater, security)	CEDIA (Custom Electronic Design & Installation Association) 2024 industry survey
Smart home market size: \$121 billion (2024), projected \$370 billion (2032), CAGR 15.4%	Fortune Business Insights, 2025
72% of luxury homeowners carry homeowners insurance with specialty carriers (Chubb, AIG, PURE) vs. 8% of standard homeowners	Insurance Information Institute
Average high-value home insurance premium: \$5,000-\$15,000/year	III; carrier rate filings
Home generator market: \$4.5 billion (2024), projected \$8.2 billion (2032), CAGR 7.8%	Grand View Research, 2025 — generator adoption signals resilience awareness

3.3 Global Residential Construction

Region	Annual New Housing Units	% with Electrical Systems	Source
North America	1.8 million	99.9%	Census Bureau; CMHC (Canada)
European Union	2.2 million	99.9%	Eurostat
China	8-12 million	99% (urban)	National Bureau of Statistics of China
India	3-5 million	96% (urban); 75% (rural, growing)	Ministry of Housing and Urban Affairs
Japan	850,000	99.9%	MLIT (Ministry of Land, Infrastructure, Transport and Tourism)
Australia/New Zealand	230,000	99.9%	ABS; Stats NZ
Global Total	~40 million (electrified economies)	—	Aggregated from national statistics

SECTION 4: MARITIME FLEET DATA

4.1 Global Commercial Fleet

Vessel Type	Number of Vessels	Average Age (Years)	Average Value (\$M)	Source
Container ships (>10,000 TEU)	~5,500	12.3	\$150-\$250	UNCTAD Review of Maritime Transport 2025
Bulk carriers	~13,000	11.9	\$30-\$80	UNCTAD 2025
Oil tankers	~8,000	14.2	\$50-\$120	UNCTAD 2025
Chemical tankers	~3,000	13.5	\$30-\$70	UNCTAD 2025
LNG carriers	~700	10.8	\$200-\$300	UNCTAD 2025
Passenger/cruise ships	~400	22.4	\$500-\$1,500	UNCTAD 2025; CLIA
Offshore supply vessels	~5,000	18.7	\$20-\$50	Clarksons Research
Ro-Ro/vehicle carriers	~2,500	16.2	\$50-\$150	UNCTAD 2025
General cargo	~17,000	22.6	\$5-\$25	UNCTAD 2025
Tugboats and workboats	~20,000	25+	\$2-\$10	Clarksons Research
Total	~105,000	22.4 avg	—	UNCTAD 2025

4.2 Maritime Fleet Newbuild Activity

Year	Newbuild Deliveries (Vessels >100 GT)	Newbuild Value (\$B)	Source
2023	2,350	\$132	Clarksons Research
2024	2,500 (est.)	\$145	Clarksons Research
2025 (est.)	2,650	\$160	Clarksons Research
2026-2030 (forecast)	15,000-18,000	\$800-\$1,200	Clarksons Research; UNCTAD

4.3 Maritime Insurance Market

Metric	Value	Source
Global marine insurance premiums (2024)	\$30.5 billion	IUMI (International Union of Marine Insurance) 2025
Hull insurance premiums	\$8.1 billion	IUMI 2025
Cargo insurance premiums	\$18.9 billion	IUMI 2025
P&I (Protection & Indemnity) premiums	\$3.5 billion	IUMI 2025
Average hull loss ratio (5-year)	68%	IUMI 2025
Average insured value — large container ship	\$150M-\$250M	Lloyd's List Intelligence
Average insured value — bulk carrier	\$30M-\$50M	Lloyd's List Intelligence
Average cargo value — 20,000 TEU container ship	\$200M-\$500M	Lloyd's List Intelligence; UNCTAD
BIMCO: 73% of total electrical failure incidents result in salvage claims >\$10M	BIMCO, 2023	

SECTION 5: INSURANCE MARKET DATA

5.1 Global Insurance Market Size

Metric	2024	2025 (Est.)	Source
Global insurance premiums	\$7.20 trillion	\$7.60 trillion	Swiss Re sigma 04/2024
Global non-life (P&C) premiums	\$4.38 trillion	\$4.62 trillion	Swiss Re sigma 04/2024
US P&C direct written premium	\$882 billion	\$940 billion	NAIC 2024
US commercial property DWP	\$120 billion	\$128 billion	NAIC 2024
US homeowners DWP	\$145 billion	\$155 billion	NAIC 2024
Global reinsurance capital	\$670 billion	\$700 billion (est.)	Guy Carpenter, 2024
Lloyd's market gross written premium	£55 billion (~\$70 billion)	£58 billion (~\$74 billion)	Lloyd's Annual Report 2024
ILS market outstanding	\$107 billion	\$115 billion (est.)	Artemis, 2024
Catastrophe bond outstanding	\$46 billion	\$50 billion	Artemis, 2024
Nat cat insured losses 2023	\$108 billion	—	Swiss Re sigma, 2024
Nat cat insured losses 2024	\$135 billion	—	Swiss Re sigma, 2025
US P&C industry surplus	\$1.04 trillion	\$1.08 trillion (est.)	NAIC 2024

5.2 Carrington Event Loss Estimates

Source	Year	Estimated Loss (\$B)	Components Modeled
National Academy of Sciences	2008	\$1,000-\$2,000	US grid replacement, GDP disruption
Lloyd's of London	2013	\$600-\$2,600	North American grid; 20-40 million people without power for 1-2 years
Cambridge Centre for Risk Studies	2016	\$500-\$2,500	GDP-at-risk modeling; 5-year recovery timeline
Swiss Re	2025	\$3,400-\$7,800	Updated with supply chain cascade, electrification growth, cyber-physical system interdependencies
AIR Worldwide (internal)	2021	\$1,500-\$4,000	Unpublished; referenced in industry discussions

5.3 Space Weather in Actuarial Models

Finding	Source
No major catastrophe model vendor (RMS, AIR, CoreLogic) offers commercial space weather model	CSM review of vendor product offerings, 2025

Finding	Source
Lloyd’s Realistic Disaster Scenario (RDS) exercise includes space weather: “A severe geomagnetic storm causing widespread disruption to electricity distribution in North America”	Lloyd’s RDS documentation, 2024
Willis Towers Watson survey: 68% of risk managers rated solar/space weather as “significantly under-modeled” vs. other cat perils	WTW Emerging Risks Survey, 2024
NAIC Catastrophe Risk Subgroup identified space weather as “emerging peril” requiring regulatory attention	NAIC Cat Risk Subgroup minutes, 2024
51% of US commercial property policies are “silent” on EMP/CME (neither explicitly covered nor excluded)	CSM analysis of ISO and carrier-specific policy forms, 2025

SECTION 6: GOVERNMENT INFRASTRUCTURE SPENDING

6.1 US Federal Infrastructure-Related Spending

Program / Agency	FY2024 (<i>B</i>) FY2025(B)	Source
DOD MILCON (Military Construction)	\$17.4	\$18.1 DOD Comptroller
DOD Facility Sustainment, Restoration, Modernization	\$28.6	\$29.8 DOD Comptroller
DOE Grid Resilience Programs (total)	\$5.2	\$6.8 DOE Budget Justification
DHS CISA Infrastructure Protection	\$2.9	\$3.1 DHS Budget-in-Brief
FEMA BRIC (pre-disaster mitigation)	\$2.3	\$2.8 FEMA Budget
FEMA HMGP (post-disaster)	\$1.5	Varies FEMA
GSA Federal Buildings Fund	\$10.1	\$10.5 GSA Budget
IIJA Grid Resilience Funding (annualized)	\$5.5	\$6.2 IIJA allocation tables
IRA Grid Resilience Funding (annualized)	\$3.8	\$4.5 IRA allocation tables

6.2 State-Level Infrastructure Spending (Selected States)

State	Annual Infrastructure Budget (<i>B</i>) <i>Resilience/EMP</i> Budget(M)	Source
Texas	\$38	\$450 (ERCOT grid hardening) Texas Legislat
Florida	\$25	\$350 Florida Offic
California	\$45	\$600 California D
New York	\$30	\$400 NYS Divisio
Virginia	\$12	\$150 Virginia Dep
Illinois	\$18	\$200 Illinois Com
Michigan	\$14	\$120 Michigan Se
Washington	\$15	\$180 Washington

6.3 International Government Infrastructure Spending (Selected)

Country/Region	Annual Infrastructure Investment (\$B)	EMP/GMD Relevance
European Union (total, all members)	\$450	CER Directive creates mandatory resilience requirements
China	\$800+	Major grid investment; Belt and Road infrastructure
India	\$180	Massive grid expansion; growing electrification
Japan	\$120	High grid resilience investment post-Fukushima
South Korea	\$85	Advanced grid; semiconductor manufacturing protection
Australia	\$45	Grid resilience; southern hemisphere CME exposure
Canada	\$35	Northern latitude EMP exposure
United Kingdom	\$50	National Grid ESO GMD programs

SECTION 7: CONSUMER AWARENESS AND ADOPTION DATA

7.1 Consumer Awareness of EMP/CME Risk

Finding	Source
Estimated 23% of US adults have heard of “EMP” (electromagnetic pulse) and can roughly describe it	CSM estimated; no formal survey exists
Estimated 8% of US adults have heard of “Carrington Event” or “solar storm” in context of grid threat	CSM estimate based on media coverage of May 2024 Gannon Storm event
Google Trends for “EMP protection” shows 340% increase in search volume following May 2024 G5 storm	Google Trends data, CSM analysis
“Prepper” community: estimated 10-20 million US adults have made some EMP preparedness investment (Faraday bags, backup power)	CSM estimate based on consumer product sales data
Home generator ownership: 6.5% of US households (2024), up from 4.1% (2019) — proxy for resilience awareness	Census Bureau, American Housing Survey 2025

7.2 Technology Adoption Curve Analysis (Rogers Diffusion Model)

CSM applies the Rogers Technology Adoption Lifecycle to model EMP/CME protection adoption:

Adopter Category	% of Market	Characteristics	CSM Target Segment
Innovators (2.5%)	2.5%	Technology enthusiasts; preppers; DOD early adopters; select insurance innovators	DOD pilot programs; Swiss Re innovation team; prepper community for Aegis Home

Adopter Category	% of Market	Characteristics	CSM Target Segment
Early Adopters (13.5%)	13.5%	Visionaries; risk-aware utilities; Lloyd's syndicates; luxury home builders	Cooperative utilities; specialty insurers; regional luxury builders
Early Majority (34%)	34%	Pragmatists; adopt after proven reference installations	IOUs after 3-5 reference installations; standard commercial property insurers; mainstream builders
Late Majority (34%)	34%	Conservatives; adopt after regulatory mandate	All remaining utilities post-FERC rule; standard homeowners
Laggards (16%)	16%	Skeptics; may never adopt until forced	N/A; CSM does not target this segment

7.3 Adoption Curve Timing (CSM Estimate)

Adopter Category	Grid Hardening	Residential	Maritime	Insurance
Innovators (2.5%)	2026-2028	2027-2029	2028-2030	2027-2029
Early Adopters (13.5%)	2028-2031	2029-2033	2030-2033	2029-2032
Early Majority (34%)	2031-2036	2033-2038	2033-2038	2032-2036
Late Majority (34%)	2036-2045	2038-2048	2038-2045	2036-2045

SECTION 8: COMPARABLE MARKET GROWTH ANALOGS

8.1 Analogous Technology Adoption Curves

Technology	TAM at Launch	Time to 10% Penetration	Time to 50% Penetration	Annual Growth Rate (First 5 Years)	Relevance to CSM
Solar photovoltaic (residential)	~\$500B	15 years	25+ years	45%	Regulatory-driven; incentive-dependent; long adoption curve
LED lighting	~\$100B	8 years	12 years	35%	Technology displacement of incumbent; energy efficiency driven
Home security systems (monitored)	~\$30B	12 years	22 years	18%	Consumer safety; insurance premium-linked; recurring revenue model

Technology	TAM at Launch	Time to 10% Penetration	Time to 50% Penetration	Annual Growth Rate (First 5 Years)	Relevance to CSM
Cybersecurity (enterprise)	~\$200B	5 years	10 years	55%	Mandated by regulation and liability; closest analog to CSM's utility/government segments
Air bags (automotive)	~\$20B	7 years	15 years	40%	Safety-critical; initially optional, then mandated; now standard
Fire sprinklers (commercial)	~\$15B	10 years	18 years	25%	Safety-critical; insurance premium-linked; building code mandated
Surge protectors (whole-home)	~\$5B	8 years	20+ years	15%	Most direct consumer analog; electronics protection; slow consumer adoption

8.2 Revenue Projection Validation Against Analogs

CSM's projected Year 5 revenue of \$3.4 billion implies a market penetration of approximately 0.55% of the \$620 billion SAM reachable within the first 5 years. For comparison:

- CrowdStrike reached \$2.2 billion revenue (approximately 1.0% of the endpoint security market) in 8 years
- Tesla reached \$31.5 billion revenue (approximately 1.2% of the global automotive market) in 15 years
- SpaceX reached \$4.6 billion revenue (approximately 1.5% of the global space market) in 15 years

CSM's projected penetration is lower than all three analogs, reflecting the conservatism of the Company's market share assumptions given the longer sales cycles in utility and government markets. If regulatory mandates accelerate adoption, penetration could significantly exceed projections.

SECTION 9: METHODOLOGY NOTES

9.1 Bottom-Up vs. Top-Down Sizing

CSM's market sizing (MKT-01) uses bottom-up methodology wherever possible — counting substations, vessels, housing units, and insurance premiums and applying cost-per-unit assumptions derived from primary sources (EPRI, RSMeans, industry interviews). Top-down estimates (e.g., “3% of global insurance premiums”) are cross-checked against bottom-up figures and used primarily for validation.

9.2 Data Limitations and Cautions

1. **EMP/CME event probability:** Long recurrence intervals (150+ years for Carrington-class events) mean the empirical record is sparse. Probability estimates from Riley (2012) — 12% per decade — are based on solar physics modeling, not a large statistical sample.

2. **Adoption curve projections:** Technology adoption for non-existent product categories must be estimated from analogs. CSM's estimates are conservative vs. cybersecurity and other regulatory-driven technologies.
3. **Government spending projections:** Subject to political change. Continuation of IIJA and IRA funding levels is assumed but not guaranteed.
4. **Insurance market response:** No precedent exists for a safety technology that addresses a peril as large as Carrington-class EMP/CME. Insurer behavior is inferred from their response to earthquake, flood, and terrorism risk, but EMP/CME may trigger different behavior due to its global, correlated nature.
5. **Maritime classification society timelines:** IACS member societies are conservative, technical organizations. Timeline estimates assume standard rulemaking processes (2-4 years from initiation to rule publication). These could be accelerated by a near-miss event.

Document prepared by: Carrington Storm Motors Market Intelligence Data sources: IEA, EIA, DOE, DHS, FERC, NERC, NOAA SWPC, NASA, UNCTAD, Clarksons Research, IUMI, BIMCO, Swiss Re sigma, NAIC, Guy Carpenter, Artemis, Lloyd's of London, NAHB, US Census Bureau, Fortune Business Insights, Grand View Research, Coldwell Banker, CEDIA, III, EMP Commission, IEEE, EPRI, NERC, academic literature as cited Last updated: July 15, 2026